



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO SENGOTTAIYAN AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

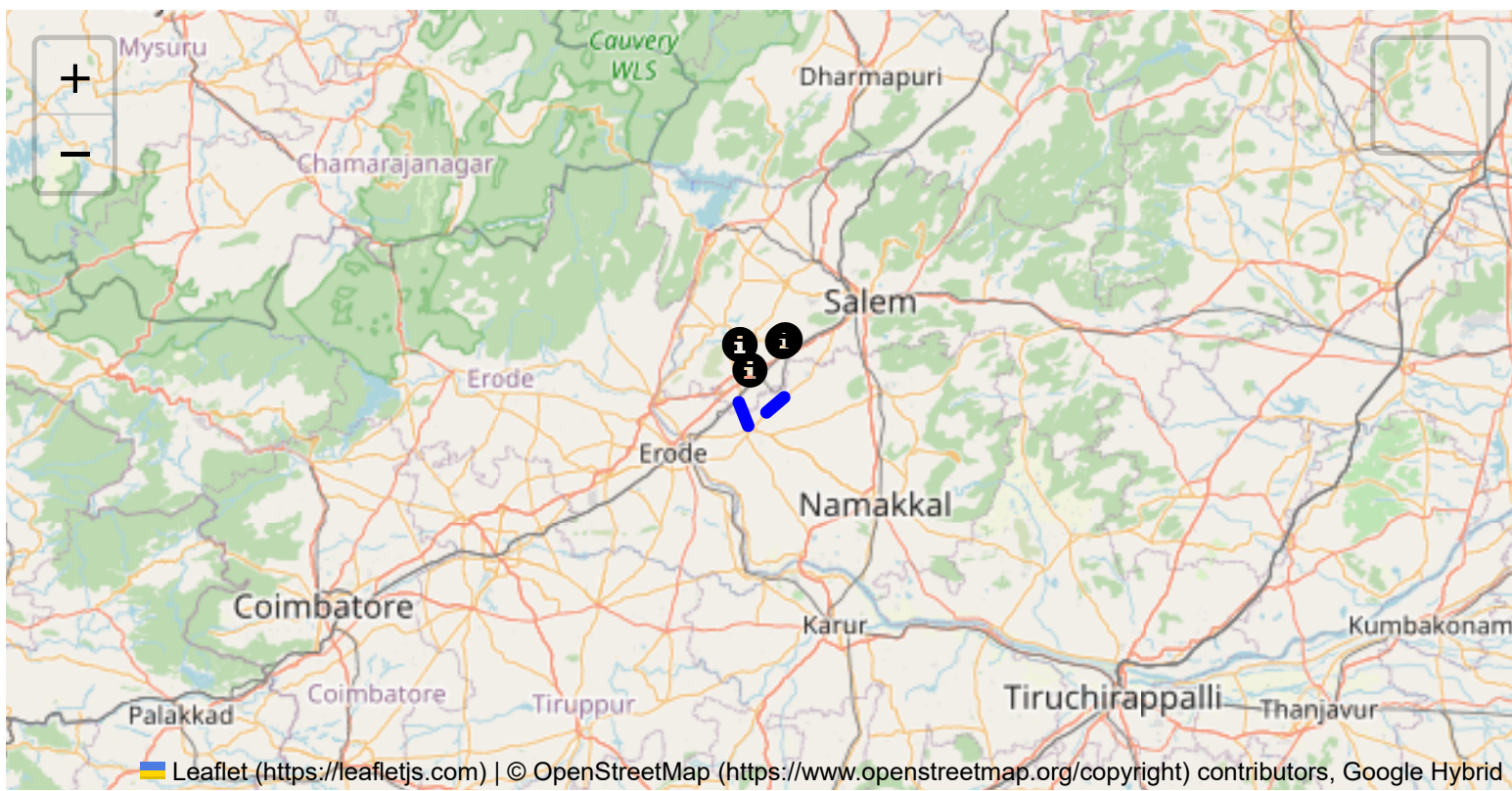
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

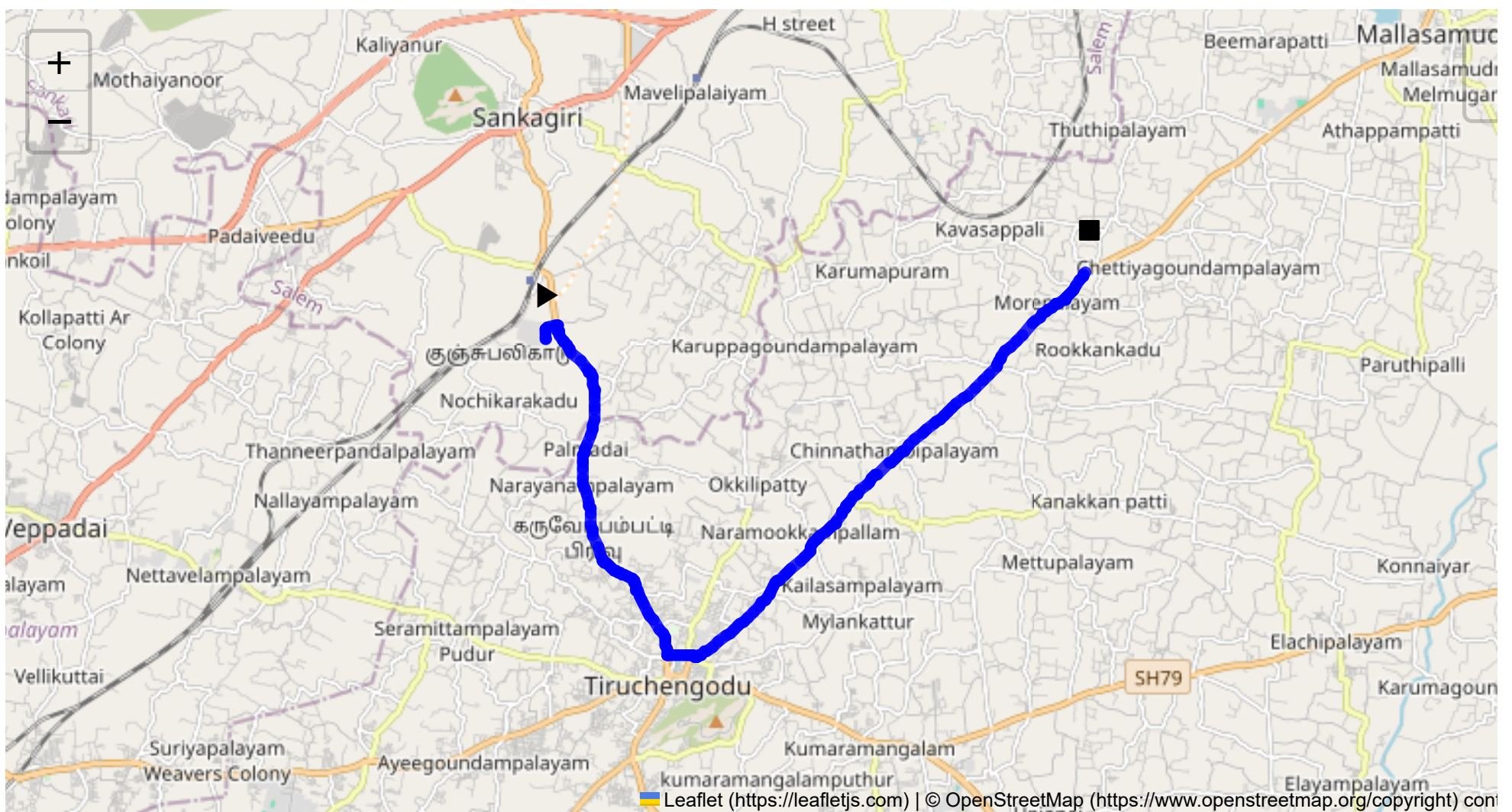
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 18.58 km
Estimated Duration: 0.5 hours
Adjusted Duration (Heavy Vehicle): 0.6 hours
Start: (11.4381, 77.8734)
End: (11.449164, 77.968147)



Welcome to the Journey Risk Management Study

Route Analysis Report

1. Overview of the Route Map

The route from Sangagiri to Tiruchengode via Puthur, the Tiruchengode Central Bus Stand, and Pillanatham stretches over approximately 18.58 kilometers. This path travels mainly along the

Tiruchengode-Salem Main Road, a major thoroughfare connecting small towns and accommodating considerable traffic, including commercial and heavy-duty vehicles.

2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu's climate varies seasonally, with hot and dry conditions typically from March to June, followed by the monsoon season from July to September, and cooler, drier months from October to February. During monsoons, heavy rainfall can cause waterlogging and reduce road visibility, leading to hazardous travel conditions. Drivers should anticipate sudden downpours and maintain caution during these months to prevent aquaplaning and avoid flooded areas.

3. Traffic Patterns and Congestion

The Tiruchengode-Salem Main Road is known for varied traffic density. Peak congestion is typically observed during mornings (7:00 AM to 9:00 AM) and evenings (5:30 PM to 7:30 PM), especially near bus stands and market areas in Tiruchengode. Traffic tends to be lighter mid-day but can still experience slowdowns due to pedestrian activities and local market zones.

4. Assessment of Road Quality and Infrastructure

The road infrastructure along this route is generally good, with well-maintained surfaces allowing smooth travel for heavy vehicles. However, some sections, particularly near smaller settlements, may have narrow lanes and require careful navigation. Periodic road work or maintenance can also create temporary diversions or reduced lanes.

5. Suggestions for Alternative Routes for Emergencies

In emergencies, detours via local rural roads may be used, although they are less ideal for heavy vehicles due to narrower paths and lesser quality surfaces. Traveling towards Erode or taking NH544 can offer alternative routes for large vehicle rerouting, should circumstances such as major accidents or road blockages occur.

6. Summary of Local Regulations Affecting Hazardous Material Transport

Transportation of hazardous materials is subject to Tamil Nadu state regulations, emphasizing safety protocols like secure loads, proper labeling, and limiting travel during peak pedestrian hours. Compliance with national safety standards (such as the use of GPS tracking) is advised to avoid legal issues and ensure public safety.

7. Overview of Historical Incidents

While major incidents involving heavy vehicles or hazardous materials are not common, there's always a risk of accidents due to vehicle malfunction or driver error. Minor collisions reported in the busier town areas stress the importance of cautious driving.

8. Environmental Considerations and Sensitive Areas

Sensitive areas along the route include school zones and residential quarters near Tiruchengode and Puthur, which require lower speed limits and heightened driver alertness. Conservation of air quality and minimal noise pollution should also be considered, given proximity to populated areas.

9. Communication Coverage and Dead Zones

Most parts of the route have reliable mobile network coverage due to proximity to urban centers. However, rural sections may experience brief coverage gaps. Equipping vehicles with satellite communication devices is recommended for uninterrupted connectivity.

10. Estimated Emergency Response Times

In towns like Tiruchengode and Puthur, emergency services respond relatively swiftly, typically within 15-30 minutes. Remote sections might face longer response times, up to an hour, necessitating prepared drivers and ready-access safety equipment.

12. Overall Summary of Risk Assessment

Overall, this route presents moderate risks, mainly affected by weather conditions and localized congestion. With good infrastructure supporting the route, it remains generally safe, provided drivers follow speed limits, remain cautious in populated zones, and prepare for adverse weather and wildlife crossings. Awareness and planning for possible communication and emergency service delays are vital for transporting hazardous materials safely.

The implementation of regular risk assessment and adherence to safety regulations will enhance safety and efficiency for all transport operators using this route.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Turn	High	11.43811, 77.87348	15 KM/Hr
2	Turn	Medium	11.43961, 77.87341	30 KM/Hr
3	Turn	Medium	11.43971, 77.87348	30 KM/Hr
4	Blind Spot	Blind Spot	11.44029, 77.87544	10 KM/Hr
5	Turn	Medium	11.38346, 77.89975	30 KM/Hr
6	Blind Spot	Blind Spot	11.44387, 77.96283	10 KM/Hr
7	Blind Spot	Blind Spot	11.44391, 77.96278	10 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
8	Turn	High	11.44387, 77.96283	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	Jayaa Hospital	11.3848223, 77.9021886	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38395, 77.89480



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.38346, 77.89975



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.44387, 77.96283



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.44391, 77.96278



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44387, 77.96283
